

Curriculum Vitae - Patrick Michael Martin II

Department of Mathematics | North Carolina State University | Raleigh, NC 27695-8205

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Research interests

Low-dimensional topology, algebraic combinatorics, homology theories for graphs and hypergraphs and their applications, topological data analysis, AI for mathematics.

Education

DOCTOR OF PHILOSOPHY IN MATHEMATICS

EXPECTED MAY 2027

- North Carolina State University
- Advisor: Radmila Sazdanovic
- Thesis: Structure of Magnitude and Chromatic Homology

MASTER OF SCIENCE IN MATHEMATICS

DECEMBER 2023

- North Carolina State University

BACHELOR OF SCIENCE IN MATHEMATICS – SECONDARY EDUCATION

DECEMBER 2018

- Mars Hill University

Preprints

- Martin II, P., Sazdanovic, R. *Torsion in Magnitude Homology Theories* (2025) available at [arXiv preprint arXiv:2503.11976](https://arxiv.org/abs/2503.11976).

Software

- Magnitude Homology, available at <https://github.com/PatrickMartinII/Magnitude-Homology.git>.
- Chromatic Homology, available at <https://github.com/PatrickMartinII/Chromatic-Homology>

Talks & Posters

- **Homology and Homotopy in the World of Graphs**, Bernoulli Center, Lausanne, Switzerland 09/26
- **Graph Based Methods in Topological Data Analysis**, Michigan State University, East Lansing, MI 07/26
- **ICTP-INdAM-SLMATH Summer School for Machine Learning**, Abdus Salam International Centre for Theoretical Physics, Trieste, Italy 06/26
- **Second Annual GRiM Symposium** UNC Charlotte, Charlotte, NC 03/26
- **TAGMaC Fall 2025** NC State University, Raleigh, NC 10/25
- **TAGMaC Spring 2025** NC State University, Raleigh, NC 03/25
- **Knots in Washington** L George Washington University, Washington D.C. 10/24
- **Richmond Geometry Meeting**, Virginia Commonwealth University, Richmond, Virginia 07/24

Funded Conferences, Summer Schools and Workshops

- **Homology and Homotopy in the World of Graphs**, Bernoulli Center, Lausanne, Switzerland 09/26

- **ICTP-INdAM-SLMATH Summer Graduate School for Machine Learning**, Abdus Salam International Centre for Theoretical Physics in Trieste, Italy 06/26
- **The Theory of Periodic Tangles and Their Interdisciplinary Applications Workshop**, Tohoku University Tokyo Electron House of Creativity in Sendai, Japan 07/25
- **Knots in Washington L**, George Washington University in Washington D.C. 12/24
- **Richmond Geometry Meeting**, Virginia Commonwealth University in Richmond, Virginia 07/24
- **Graduate Student Conference in Geometry, Topology, and Algebra**, Temple University, Philadelphia, PA 05/23

Seminars

- **Topics in Knot Floer Homology**, led by Tye Lidman Fall 2025
- **Involutive Khovanov Homology**, led by Biji Wong Fall 2025

Organized Seminars

- **Homological Algebra and Hochschild Homology** Fall 2026

Teaching Experience

AT NORTH CAROLINA STATE UNIVERSITY

AUGUST 2022 – CURRENT

- Instructor of Record | MA 114: Finite Mathematics | 50+ Students | Fall 2023, Summer 2024
- Instructor of Record | MA 241: Calculus II | 100+ Students | Spring 2024
- Recitation Leader | MA 241: Calculus III | 50+ Students | Fall 2024, Fall 2025
- Recitation Leader | MA 241: Calculus II | 100+ Students | Spring 2026

AT CLEVELAND HIGH SCHOOL

AUGUST 2019 - JUNE 2022

- Instructor of Record | NC Math I: Basic Algebra | 40+ Students | 3 Sections
- Instructor of Record | NC Math II | Algebra & Geometry | 30+ Students | 3 Sections
- Instructor of Record | NC Math III | Algebra & Trigonometry | 30+ Students | 2 Sections
- Instructor of Record | Honors NC Math III | Algebra & Trigonometry | 30+ Students

Computer Skills

- Erdős Institute Summer 2025 Data Science Boot Camp
- Python - sklearn, pandas, numpy, scipy, sagemath
- C++